EARSAFE METER V1

Battery-powered desk sound monitor for personal sound exposure control.

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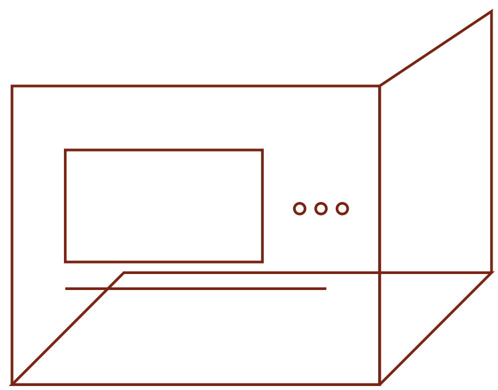
Design intent

- Measure room sound near the listener position.
- Show one clear sound level value at a glance.
- Show warning state before daily sound dose becomes excessive.
- Run from internal battery power with USB-C charge.



Front and side concept view

SECTION 00

Contents

This specification defines the first build of EARSAFE METER. The device is a battery-powered desk sound monitor. It shows present sound level, warning state, peak level, and daily sound dose.

DOCUMENT INTENT

This document must stay direct. It must stay technical. It must not use decorative filler, startup language, or sales language.

SECTION 01

Purpose and Use Limits

EARSAFE METER V1 is a desk sound monitor. The device measures sound near the user position. The device shows the present A-weighted sound level. The device also shows warning state, peak level, and daily sound dose.

Use the device for personal sound control during music use, speaker testing, long study sessions, and other repeated listening activity. Do not use the device as a certified legal meter. Do not use the device as the only control for industrial safety.

USE LIMIT

The reading becomes useful only after calibration against a known reference meter. Before calibration, use the value as relative guidance.

SECTION 02

System Summary

ITEM	TARGET	ENGINEERING REASON
Form	Self-standing desk unit	The device must sit at the listener area without extra support.
Display	2.13 inch black-and-white e-ink	The display must stay readable and low power.
Controller	nRF52840 class module	The controller must support low-power sampling and simple peripheral control.
Microphone	Digital MEMS microphone	The signal path must stay stable and simple.
Battery	3000 mAh LiPo nominal	The device must give practical battery life.
Core outputs	Current dBA, 1 min average, peak, daily dose	The user must see present state and exposure history.

- Keep the first build simple.
- Use standard modules before a custom board.
- Do not depend on Wi-Fi in the default mode.
- Make one value lead the screen.

SECTION 03 Functional Requirements

ID	REQUIREMENT	PASS CONDITION
REQ-01	The device must show present A-weighted sound level in dBA.	User can read the value on the main screen.
REQ-02	The device must show warning state at 80 dBA and alert state at 85 dBA.	State changes at the set thresholds.
REQ-03	The device must track cumulative daily sound dose with a 3 dB exchange rule.	Dose value rises during a controlled sound test.
REQ-04	The device must operate from internal battery power.	Device runs with no USB cable attached.
REQ-05	The device must store calibration offset in non-volatile memory.	Offset remains after power cycle.
REQ-06	The device must show low-battery state before shutdown.	User sees warning before the battery is depleted.

SECTION 04 Hardware Configuration

V1 should use a development-board architecture. This rule reduces risk. This rule also reduces build time.

UNIT	PREFERRED CLASS	NOTES
Main board	XIAO nRF52840 Sense or equivalent	Pick a board with battery support if available.
Microphone	On-board digital mic or external I2S/PDM MEMS mic	External mic gives better placement freedom.
Display	2.13 inch e-ink module	Do not use a tri-color display in V1.
Battery	Protected 1-cell LiPo	Use a protected cell only.
Charge path	USB-C 5 V input	Do not require USB PD for V1.
User input	One or two push buttons	Keep the interface simple.
Optional alert	Low-current LED or piezo buzzer	The alert may be omitted in the first quiet prototype.

SECTION 05 Signal and Control Architecture

The sound enters through the digital microphone. The controller reads the signal. The firmware applies A-weighting and RMS logic. The firmware then computes present dBA, 1 minute average, peak, and daily dose.

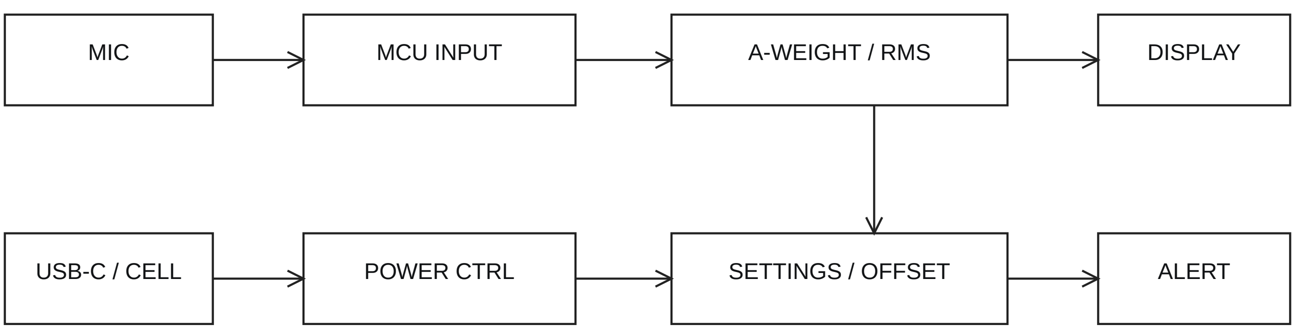


Figure 1. The signal enters through the microphone. The MCU computes the sound level. The logic updates the display, alert state, and stored calibration data.

1. Read digital microphone samples at a fixed sample rate.
2. Remove DC bias if the selected microphone path requires it.
3. Apply A-weighting filter in firmware.

4. Compute RMS energy for the update window.
5. Convert the level to dBA with the stored calibration offset.
6. Update the display only when the visible state changes or on the scheduled refresh step.

SECTION 06 Display and User Interface

ELEMENT	RULE	PURPOSE
Main level field	Use the largest type on the screen.	The user must read the sound level from across the desk.
State field	Use one short word.	The user must know the condition at once.
Dose field	Show percent of daily target.	The user must see long-duration exposure.
Peak field	Show maximum recent value.	The user must see transient high sound.
Battery field	Show icon or percent.	The user must not lose power without warning.

STATE	ENTRY CONDITION	USER MEANING
NORMAL	Below warning threshold	The sound level is acceptable for present use.
WARNING	At or above 80 dBA	The user should manage time or reduce level.
ALERT	At or above 85 dBA	Exposure control is now important.
HIGH ALERT	At or above 94 dBA	The user should reduce sound level at once.
LOW BATTERY	Battery under low limit	The device needs charge soon.

SECTION 07 Power System

The default operating mode is battery power. USB-C is for charge and optional desk power. The firmware must reduce needless display refresh. Wireless functions must stay off unless the user needs them.

PARAMETER	TARGET	COMMENT
Nominal battery	3000 mAh	This value is a starting point. Test may change it.
Charge input	USB-C 5 V	Keep the charge path simple.
Low-battery warning	20 percent target	The exact value may change after battery test.
Primary alert method	Display state change	This method uses less power than continuous active output.

SECTION 08

Calibration and Accuracy

A digital microphone does not give true dBA by itself. The device needs calibration against a known reference. V1 should use a steady test sound and a reference meter at the same physical position.

1. Put the reference meter and the device at the same point.
2. Use steady pink noise or another stable test sound.
3. Wait for both readings to stabilize.
4. Read the two values.
5. Store the difference as the calibration offset.
6. Repeat the test after a major hardware change.

ACCURACY RULE
Do not claim legal meter accuracy. After calibration, use the value as a personal hearing-safety guide.

SECTION 09

Mechanical Arrangement

The case should be a compact wedge or upright desk form. The front face should point to the user. The microphone port must have a direct acoustic path to room sound.

FEATURE	REQUIREMENT	DESIGN RULE
Case form	Self-standing body	Do not require a separate stand.
Display angle	Readable at normal desk view	Bias the face to seated eye height.
Microphone opening	Direct acoustic port	Do not block the port with dense structure.
USB-C access	Rear or side entry	Do not force a sharp cable bend.
Battery bay	Protected internal space	Keep the cell away from sharp edges and screws.

SECTION 10

Prototype Build Sequence

1. Make the electronics work on the bench.
2. Show a basic e-ink screen.
3. Read live sound data from the microphone.
4. Compute present dBA and daily dose.
5. Set threshold logic and state labels.
6. Calibrate the device with a reference meter.
7. Move the system into the printed case.
8. Repeat the calibration and battery tests after final assembly.

SECTION 11

Verification Tests

TEST ID	TEST	EXPECTED RESULT
T-01	USB-C charge test	Battery charges and the device stays stable.
T-02	Battery-only run test	Device runs with no USB cable attached.
T-03	Sound response test	Level value changes with controlled source level change.
T-04	Threshold test	State changes at the configured thresholds.
T-05	Dose accumulation test	Dose rises with controlled exposure duration.
T-06	Calibration retention test	Offset remains after power cycle.

SECTION 0A

Appendix A Preliminary Bill of Materials

REV A

LINE	ITEM	QTY	NOTES
1	Main controller board	1	nRF52840 class board with digital microphone support.
2	E-ink display module	1	2.13 inch black-and-white module.
3	Protected LiPo cell	1	3000 mAh nominal starting target.
4	Push button	1-2	For screen change and calibration mode.
5	USB-C cable or receptacle path	1	For charging and optional desk power.
6	Printed enclosure	1	Two-part shell with microphone port and display window.
7	Optional LED or buzzer	0-1	Use only if the warning mode needs it.